Asad Ikram

Computational social science · measurement validity of machine-labelled text · large-scale social data collection

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Research interests

Whether the instruments computational social science uses to measure public opinion actually measure the same thing across platforms, across model versions, and across the standpoints built into the models. I approach this as a measurement problem, using psychometric tools (generalizability theory, measurement invariance and DIF, score linking) on machine-labelled text, and I bring seven years of building the production annotation pipelines whose failures motivate the question.

Publications and research output

Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment *arXiv, cs.SI*

Asad Ikram and Dr Ella Haig (University of Portsmouth) · 23 pages · ICWSM 2027 submission planned

- Treats each social platform as a **survey mode** and specifies a six-step psychometric protocol for machine-labelled sentiment: generalizability theory variance decomposition, measurement invariance and differential item functioning testing, and score linking with policy events as anchors.
- Known-truth simulation, **500 replications**: a pooled cross-platform index that takes platform scores at face value reports the **wrong sign of aggregate sentiment on roughly a quarter of days**, purely from measurement differences. A measurement-adjusted index does not.

Calibrating the Observatory: Measurement Equivalence, Instrument Drift and Epistemic Standpoint in LLM-Annotated Public Opinion *Doctoral proposal, 2026*

Full proposal and reference list at asadfix.github.io/helsinki_proposal

- Four studies on one data spine. **Platform equivalence** of LLM labels tested with fitted categorical invariance and DIF models with a human criterion inside the measurement model; **instrument drift** handled by a shadow-anchored bridging design across an executed open-weight version ladder, so a longitudinal series stays comparable when the annotator changes underneath it; **epistemic standpoint** as a fixed facet in a multivariate generalizability design; and a criterion test against probability-sample surveys.
- Deliverables are an open toolkit and an **Instrument Change Log**, a reporting standard for annotation pipelines.

MSc Dissertation: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies *2025*

University of Portsmouth · Distinction (mark 78) · Supervisor: Dr Ella Haig · ethics approval TETHIC-2025-111094

- Collected and integrated **279,000 public posts across Twitter/X, Reddit, YouTube and Quora** on UK economic policy via official APIs and licensed feeds, under GDPR-compliant procedures.
- Fine-tuned **BERT at macro-F1 0.878** against BiLSTM and lexicon baselines; human validation on a balanced gold set (Cohen's kappa ~0.78); temperature-scaling calibration; McNemar and bootstrap significance testing; Prophet forecasting of the daily sentiment index; NRC emotion analysis; interpretability via LIME and integrated gradients.
- Central finding: the same policy event produced **structurally different sentiment and engagement profiles on each platform**, which is the divergence the methods paper then formalises.

web-scraping-guide.com *Public reference, maintained*

- A free 24-section engineering reference on web data collection: TLS and JA4+ fingerprinting, anti-bot systems, proxy strategy, the legal landscape (hiQ v. LinkedIn, Van Buren, Meta v. Bright Data), and the silent-failure modes of collected data, with a 130-node knowledge graph published as machine-readable context.

Research-relevant professional experience

Founder and CTO, ArtemisAI Ltd *Feb 2025 – present*

UK company · social media intelligence platform

- Designed and operate a multi-model annotation cascade over **3.39 million comments**: six NLP endpoints (sentiment, emotion, toxicity, topic, language, intent) with a confidence-routed LLM fallback tier, plus CLIP image-text fusion, OCR and speech transcription for video.
- The measurement finding that motivates my doctoral work: our two model tiers agree on **82.9% of sentiment labels and 0.0% of toxicity labels**, while the toxicity model reports **96.7% average confidence**. The dashboard's "accuracy" figure is, on inspection of the code, inter-model agreement with no criterion behind it. The zero comes from mismatched label taxonomies between tiers, which is an equivalence failure of exactly the kind this research addresses.
- Built the honesty controls in response: a QA layer that reports "not measured" and "not comparable" rather than a fabricated zero, per-dimension agreement tables, and escalation share as a model-health signal.

Data Engineer Consultant, M+C Saatchi Fluency
Dec 2024 – present

UK, remote · social listening infrastructure for UK Government departments, Amazon, Ford, Nike and Reckitt

- Operate the multi-platform ingestion pipeline behind campaign and policy intelligence: licensed feeds plus custom crawlers into AWS, with NLP sentiment models; reports are read by senior government communications teams.
- Built a **self-healing crawler fleet (68 crawlers)** with an agentic repair system, and the data-quality layer around it: **KL-divergence drift alarms** on field distributions, per-column coverage alerts, dedup registries, and post-crawl verification of every run. Maintain documented filter rules, permanent removal lists and audit logs on government-facing datasets.

Senior Data Engineer, Dubizzle Labs
Feb 2022 – Aug 2024

Pakistan · promoted twice in two years (Associate SWE ⟶ Data Engineer ⟶ Senior)

- Led a three-engineer team owning **500+ regional web scrapers across fifteen countries**, collecting **multilingual data in English, Arabic, Urdu and Hindi** for market and consumer-behaviour analytics. Multilingual collection at scale in Global South information environments is routine work rather than a project risk for me.
- Redesigned the layered data warehouse, improving ETL throughput 60%; introduced unit testing and crawler contract validation, cutting production collection incidents 45%.

Data Engineer Consultant, Fix.com
May 2023 – present

Canada, remote, concurrent

- Pipelines processing **50M+ data points from 200+ sources**; built an 86-crawler control plane with streaming data-quality gates, coverage validation against site-reported truth, and evidence-based proxy tier selection.

Earlier consulting
2018 – 2026

- VendueTech (Ireland, 2025) · Prefe (Italy, 2024) · CXG (Dubai, 2025–26) · Senior Freelance Data Engineer, 300+ projects across 400+ websites for 50+ clients (2018–2026).

Education

MSc Data Analytics — Distinction
University of Portsmouth, UK · 2024–2025

- Intelligent Data and Text Analytics; Big Data and ML; Explainable AI; Data Management (80).

BSc Computer Science
FAST-NUCES, Pakistan · 2017–2022

Awards

- **Chevening Scholar 2024/25** — the UK FCDO's flagship international scholarship, awarded to roughly the top 2 to 3% of more than 70,000 applicants across 160 countries.

Methods and tools

Measurement	Generalizability theory (incl. multivariate G-studies), multi-group and ordered-categorical CFA, measurement invariance testing and DIF with anchor purification, score linking and equating, MTMM, Monte Carlo simulation with known ground truth.
NLP and ML	Transformer fine-tuning (BERT), BiLSTM, LLM annotation and distillation pipelines, calibration (temperature scaling, isotonic), interpretability (LIME, integrated gradients, SHAP), CLIP embeddings, Whisper, Prophet.
Statistics	McNemar tests, bootstrap confidence intervals, inter-annotator agreement (Cohen's kappa), stratified probability sampling, KL-divergence drift detection.
Data collection	Multilingual collection at scale (English, Arabic, Urdu, Hindi), official platform APIs (Meta Graph, Reddit, YouTube), licensed feeds, Scrapy and Playwright, ethics and GDPR-compliant procedures.
Engineering	Python, PySpark, SQL, AWS (Fargate, Glue, Athena, Redshift, SageMaker), Airflow, Docker, Kubernetes. **Languages:** English (fluent), Urdu (native).